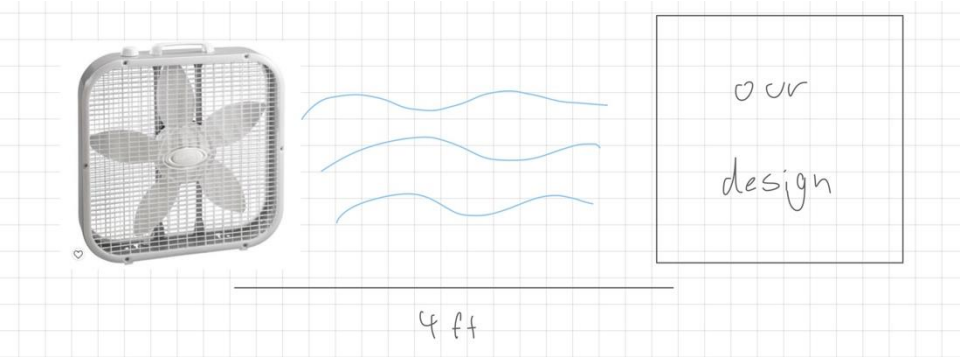
Cooper Smith, Kasin Ito-Sato, Isaac Sorensen

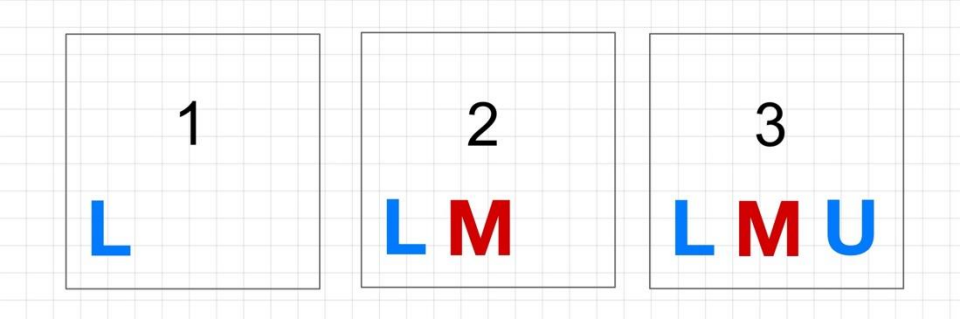
Conceptual Design Report

Design Requirements and Functions:

Input: Catch the incoming air from a box fan located 4 feet away and use this to trigger the action of your segment. Be careful that the action does not trigger from ambient indoor airflow.



Output: One at a time, reveal the letters “L”, “M”, and “U” as part of a cinematic finale. You will need to construct the letters in whichever fashion you choose.

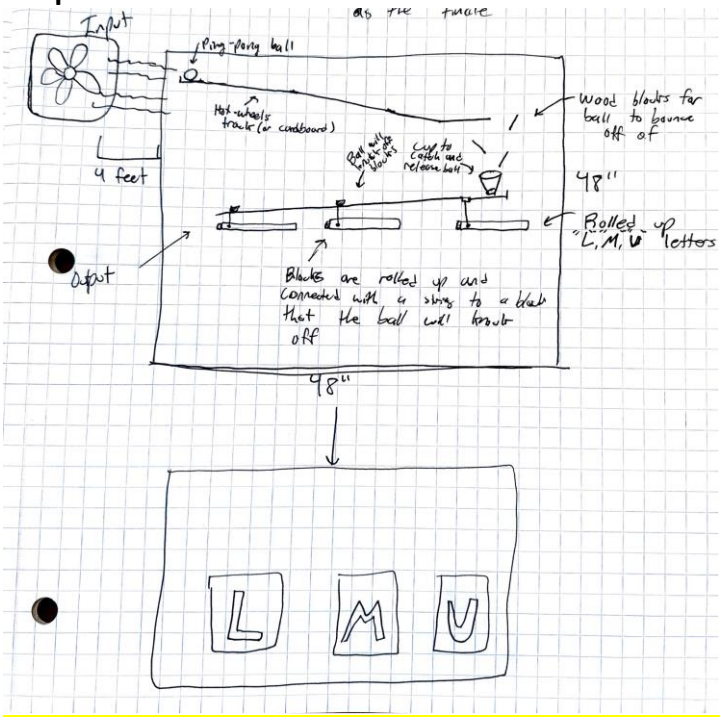


Design requirements:

- Takes an input from a fan 4 feet away that starts the rest of our design
- Ends in the letters spelling LMU to reveal
- Must be fully functional and reliable without any stops or breaks in the system
- Total design must be 4 feet wide and 4 feet tall
- Must be created within the given time frame and be completed by our end date
- Must include multiple different parts or small systems on our main design

Design Alternatives (Individual Portion):

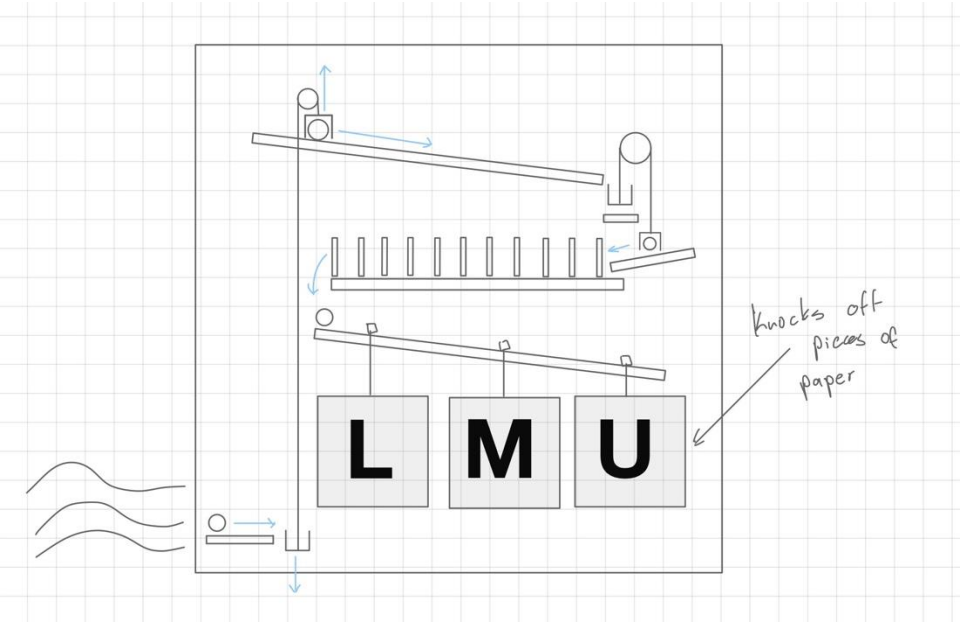
Cooper:



The fan as the input should be on the ground, not in the air as shown in the picture. I did not realize this as it was not specified in the description for the input, so I assumed it was at the top when I did my drawing. We only later figured out it should be on the bottom after asking in class when I already completed and turned in my drawing. To make the input go from the bottom to the top to make my design better fit the criteria, I could have added a large upright lever that a ball on a track with the fan input runs into which in turn pulls a string away from the top track releasing the ball at the start and completing my design.

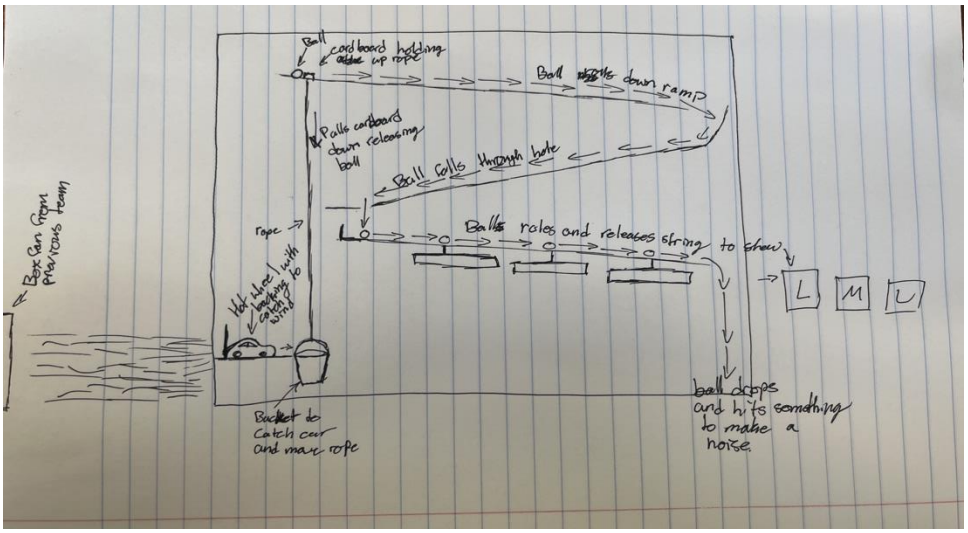
As seen with my design, it starts with the fan input to trigger a ping-pong ball on a track that is attached to the 48” by 48” plywood board. I then have this ping-pong ball hit three different wood blocks to bounce off of and add some more effect to the final design. This ball then funnels into a cup with the bottom cut out, to catch the ball, landing on another track after. This track has 3 separate blocks in the way, each of which are attached to one of the three rolled up LMU letters. Knocking off each block will make each LMU letter unroll unveiling the final output we were assigned. The bottom part of my sketch shows how I plan for the LMU letters to roll out at the end of our design. I prioritized the overall functionality and how reliable my design it. Without a reliable, consistently working design it would not fit in with the rest of the class and there would be many breaks in the entire system. This would make for a less pleasurable viewing experience and take away from the overall project. Making sure this all fit on our provided 48” by 48” plywood sheet was an important prioritization as well.

Kasin:



My design takes the input from the fan on the floor four feet away by pushing a ball into a cup. This cup is attached to a pulley that when weighed down on one side, will release another ball at the top of the 48” by 48” plywood sheet. This ball will roll down and affect another similar pulley system. This pulley will release another ball hitting a block starting a domino effect. The domino effect will then release another ball that will roll along a slanted ramp and knock off hooks holding pieces of paper to reveal the letters “L”, “M”, and “U”. Throughout my design I prioritized functionality the most when creating my design. I wanted to include parts that would consistently work and did not rely on chance. The pulleys and ramps will most likely work consistently. The only part of my design that may have a chance of malfunctioning is the domino portion they either fall off or do not fall.

Isaac:



My design starts with the box fan from the end of the previous segments design pushing a hot wheel's car with cardboard on the back dropping into a cup. The cup is connected to a 3.5-foot rope/string that attaches to a piece of carboard, when the cardboard drops it releases a ball. The ball rolls down two ramps then will drop through a hole. This hole leads to one last ramp with 3 pieces of string attached to rolled up paper. As the ball passes through the string, it releases the roll in the paper, revealing LMU as the result. All of the ramps, strings, signs, car, and balls will all be attached to a 48" by 48" sheet of plywood. The output was my main focus in this design. I had to figure out a way that the input would result in our designated output using all of the design requirements as well as having a little bit of creativity and surprise for the audience. For my design I prioritized the pulley system at the beginning of the machine as well as the reveal of the LMU lettering. I mainly focused on a way that our machine would start and finish smoothly every time without fail. I did not want to have the machine be too complicated where there would be more possibility for error, but I also made sure to integrate interesting factors to the final design. For my design I believe that that the reveal of the letters has the most opportunity for fail. It has a chance of either stopping the ball from continuing to the next letter or it has the ability to not unroll at all.

Design Decision Matrix:

Design:	Functionality	Tempo	Surprise	Comedy	Total
Cooper	2	7	8	9	26
Kasin	8	8	9	9	32
Isaac	8	6	8	9	31

Design Analysis:

The scores for functionality were based on whether the different attributes of our designs would work effectively or not. The scores between our designs were similar because there is a general level of simplicity to our designs which will allow for strong functionality. The tempo is based on how entertaining the design would be to watch, all our designs are generally similar which will create a solid tempo. The surprise factor is one that can be hard to achieve along with functionality. We hoped to have a design with some surprises to add interest to which we considered our final design choice. Finally, the comedy aspect is the main point of entertainment. Comedy can be a hard factor to include in such a simple design, but our design all has similar ways to add some comedy along with surprise.

For Cooper's design, we chose a low score for functionality since the fan input started at the top instead of the bottom of design. This would make it nonfunctional, and some adjustments would have to be made. We only realized this after asking for clarification. There isn't a huge amount of complexity to it and not much that would go wrong in the design. We added a surprise factor of 8 for the letters unveiling in the way they do. In terms of tempo, we had Cooper's design at a 7. The tempo wasn't necessarily fast but wasn't slow either. We chose it to be on the side of better tempo after comparing to the other designs. The way the ball went falls on the tracks and goes through the system is what mainly contributed to this score. For this reason, Kasin also has a high surprise factor. Lastly, we put comedy at a level 9 partly because of the letters falling but also because how the ball on the tracks knocks off the blocks to reveal the letters. This gives a total score of 26.

For Kasin's design, we chose a score of eight for functionality as the concept should work effectively but some features have a chance of malfunctioning. The features that may malfunction are most likely the domino aspect as they may fall off the design or not be placed a proper distance apart and not fall over. We then chose a score of seven for tempo because there are a good number of moving parts but also areas that could be considered slow. The areas that may be slow are the domino and LMU letter reveal as the dominoes may not fall fast and the letter reveal depends on the velocity of the ball down the ramp. The complexity of the design, however, will dry interest and anticipation for the next event to occur. We then gave a score of eight for surprise because of the pulleys and the reveal of the LMU letters. The pulleys reveal a ball which adds some surprise but the reveal of the LMU letters could be improved and made more. Finally, for comedy, we gave a score of nine because of the many moving parts that add to the overall enjoyment of the system.

For Isaac's design we chose a score of 8 for functionality because the design meets the design requirements and for the most part it will work well, one concern that we had was that we were unsure how to make the bucket and string mechanism work. With our added surprise category Isaac's design scored an 8 due to the randomness of the ball dropping from ramp to ramp as well as the noise it would make at the end. In the last category, Isaac's design scored a 9 because the use of the ending noise and the random ball droppings adds a little bit of a comedic effect to the design. Isaac's design scored a 6 in the "tempo" section of the design matrix. This is

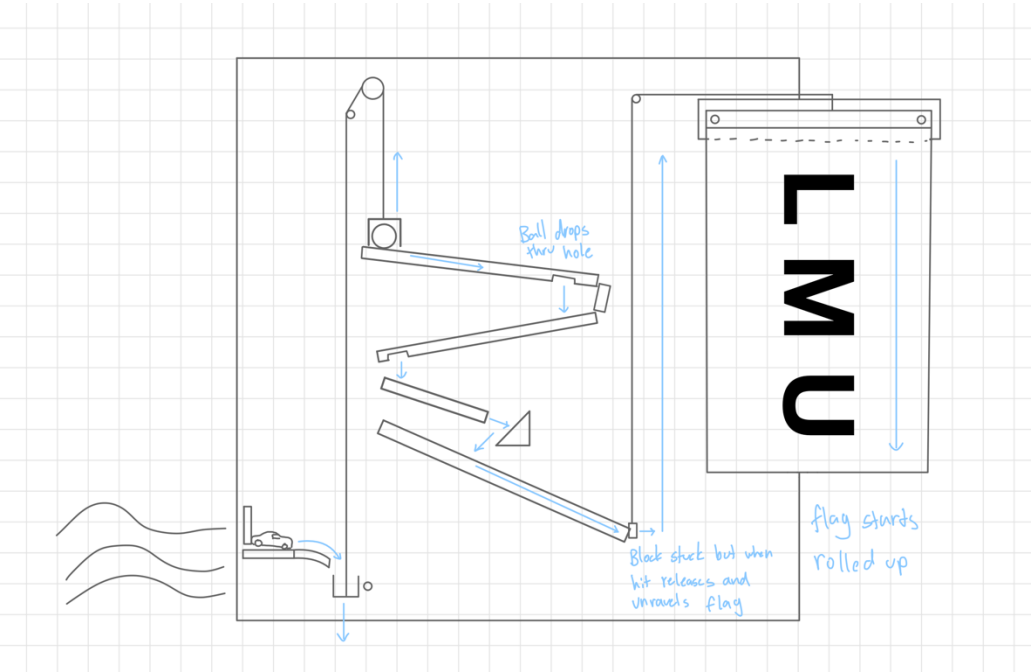
due to the randomness of the ramps and ball dropping. Isaac's design had a total score of 31 in the design decision matrix.

Design Selection:

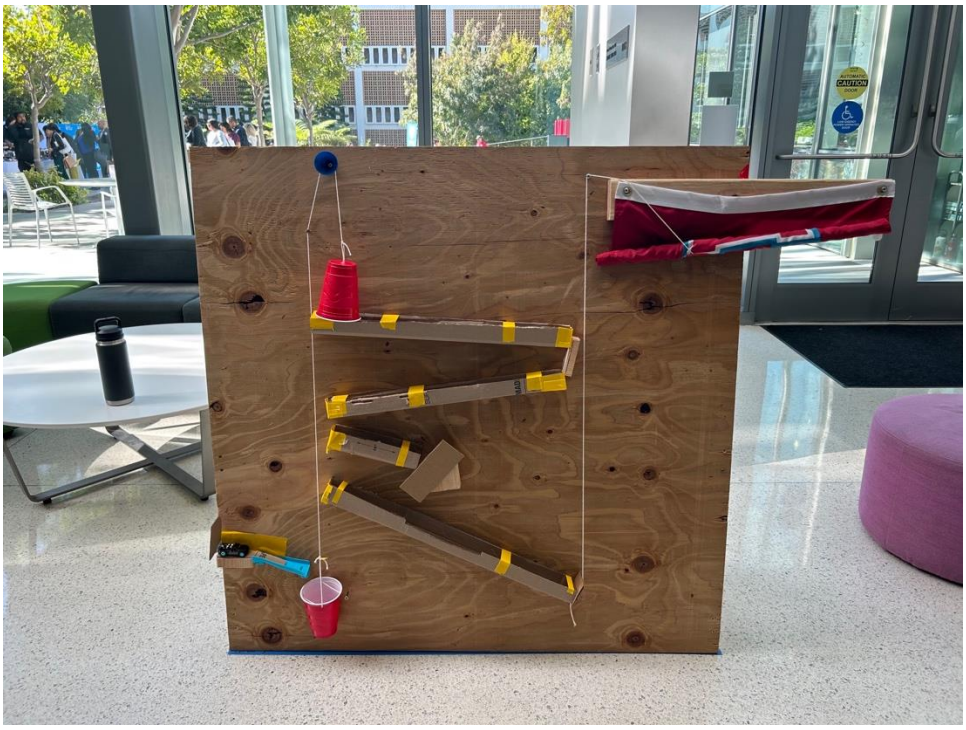
We chose to go with a design like Kasin's design. All our designs were somewhat similar, but Kasin's seemed to be a combination of our other two designs and had a generally good scoring in each of our categories. We are not tied down to following this design exactly and want to continue to workshop what could add more interest and value to our design. One of the places we think we should think further about is the reveal of the LMU letters. This wasn't the deciding factor, but he also received the highest score which helped influence us to pick his design. We liked his tempo and overall complexity, shown in our scores, better since it was just laid out better and an interesting design, we think would keep people entertained. The main difference between our designs was the section between the input and output. Isaac and Kasin had a similar idea to use a pulley as the input as it would allow us to use gravity to our advantage and Cooper agreed with this idea. Throughout the middle section of our design, we all used ramps but in Kasin's he also included a domino section that may add some interest. Isaac's design also had a slightly longer ramp section than Cooper's design. Cooper and Isaac both had a similar idea to have the output as the LMU letters falling onto the board, while Kasin's had pieces of paper covering the letters that would be knocked off.

Design Implementation:

Final Sketch:



Actual Design:



Our design starts with an input from a box fan four feet away and the wind from this fan is caught by a car with a large rectangular piece of cardboard attached to its back. This car then falls into a cup which activates a pulley that releases a large metal ball down a set of ramps. The ramps go back and forth down our design and have holes where the ball falls through or a backboard that the ball bounces off of into the next ramp. These ramps then lead to the bottom of the board and the ball ends up hitting a piece of cardboard attached to a string that is lightly taped to the last ramp. This string is holding the rolled-up LMU flag and when the cardboard piece is hit, it gets dislodged from the ramp and the flag is released revealing the LMU letters one by one. The flag has a pole attached to the bottom which allows it to be heavy and roll down nicely.

Design Evaluation:

Our design was successful in completing the task of using the box fan as input and having the LMU letters be revealed one by one at the end. Our input using the pulley and car activation was able to work consistently as predicted in our analysis. One of the first challenges we faced was ensuring the ball was kept on the ramps. There were times when the ball would hit an edge or roll weirdly and would fly off the board, but we combatted this by adding extra borders and pieces of wood to ensure it would stay on track. Our largest challenge was figuring out a consistent way to have the flag roll down. We tried many different methods like adding screws as contact points, using pipe cleaners so the string moved smoother, and even using the rim of a red solo cup in hopes the piece of cardboard could fit through it and allow the flag to fully roll down. We eventually found a consistent method that only needed one screw placed strategically above the end of the last ramp. This allowed the cardboard piece to not get stuck

on anything as well and only having one contact point meant the string would move smoothly and release the flag effectively.